

PROJECT REPORT

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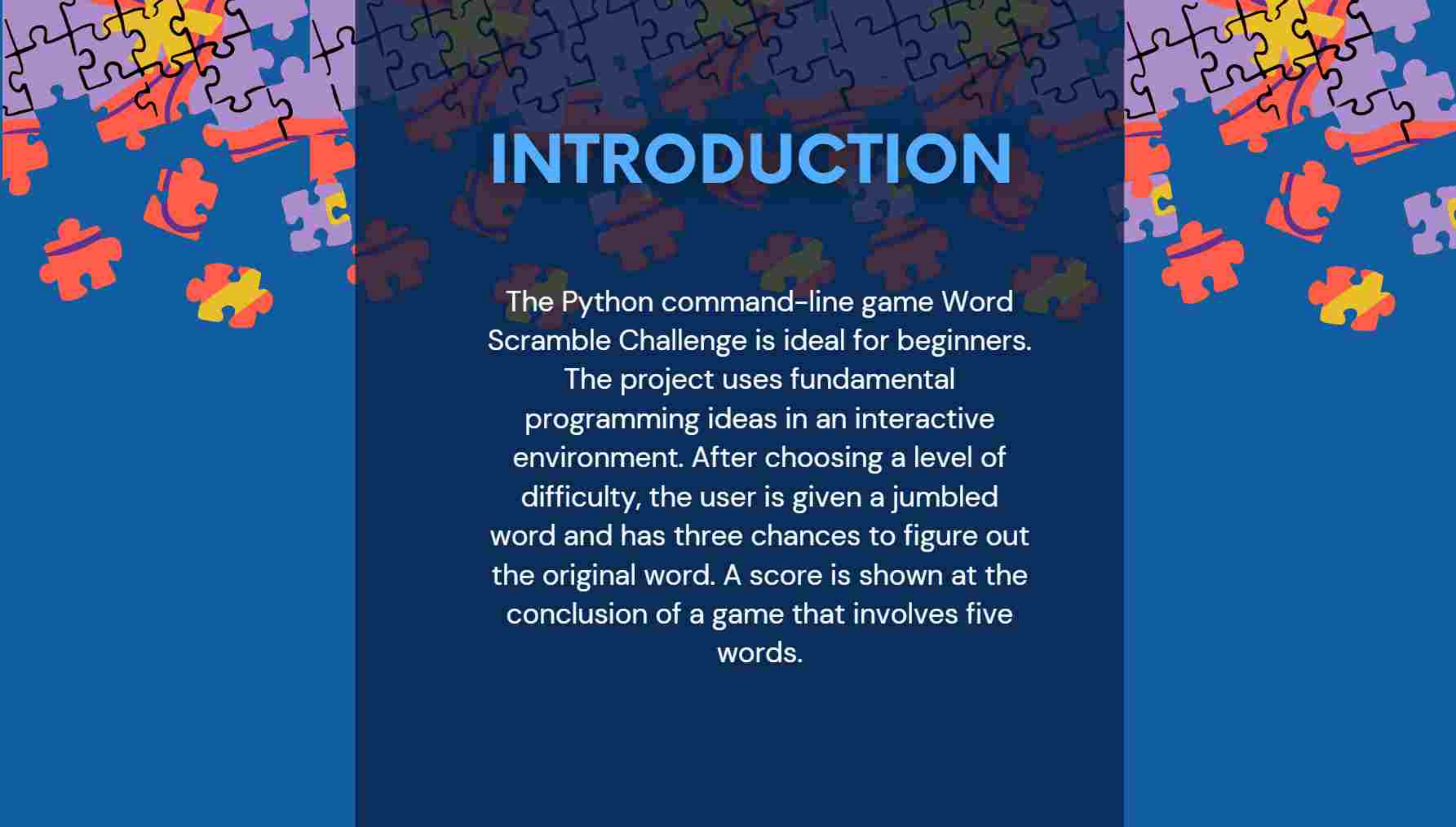
Project Title: Word Scramble Game

Course Subject: Python Programming & Problem Solving

Repository Link:

https://github.com/srishtimaurya23/Word_scramble_Game.git





INTRODUCTION

The Python command-line game Word Scramble Challenge is ideal for beginners.

The project uses fundamental programming ideas in an interactive environment. After choosing a level of difficulty, the user is given a jumbled word and has three chances to figure out the original word. A score is shown at the conclusion of a game that involves five words.

PROBLEM STATEMENT AND OBJECTIVES



PROBLEM STATEMENT

Practicing programming ideas solely through theoretical exercises might be challenging for beginners. This project offers a brief interactive application that allows Python concepts like dictionaries, lists, functions, loops, conditions, random selection, and input validation to be applied to a real-world issue.



OBJECTIVE

- Using Python, create a working command-line word game.
- Using distinct source files, implement modular programming.
- To make each game unique, use random selection.
- Give limited attempts and varying degrees of difficulty.
- Verify typical user input.
- Determine and show the player's score.
- Use Python's built-in unittest module to demonstrate testing.



FUNCTIONAL REQUIREMENTS

01

Word Management:
adds words to the game
and keeps them in the
Easy, Medium, and Hard
categories.

02

Game logic: chooses five
words, jumbles them,
controls attempts and
rounds, verifies
responses, and
computes the score.

03

User Interaction: shows
the game's interface,
takes input for difficulty,
response, and replay, and
transmits outcomes.

04

Score Management:
shows the total number
of games, average score,
and total score.

05

Input Validation:
verifies and adjusts
user input.



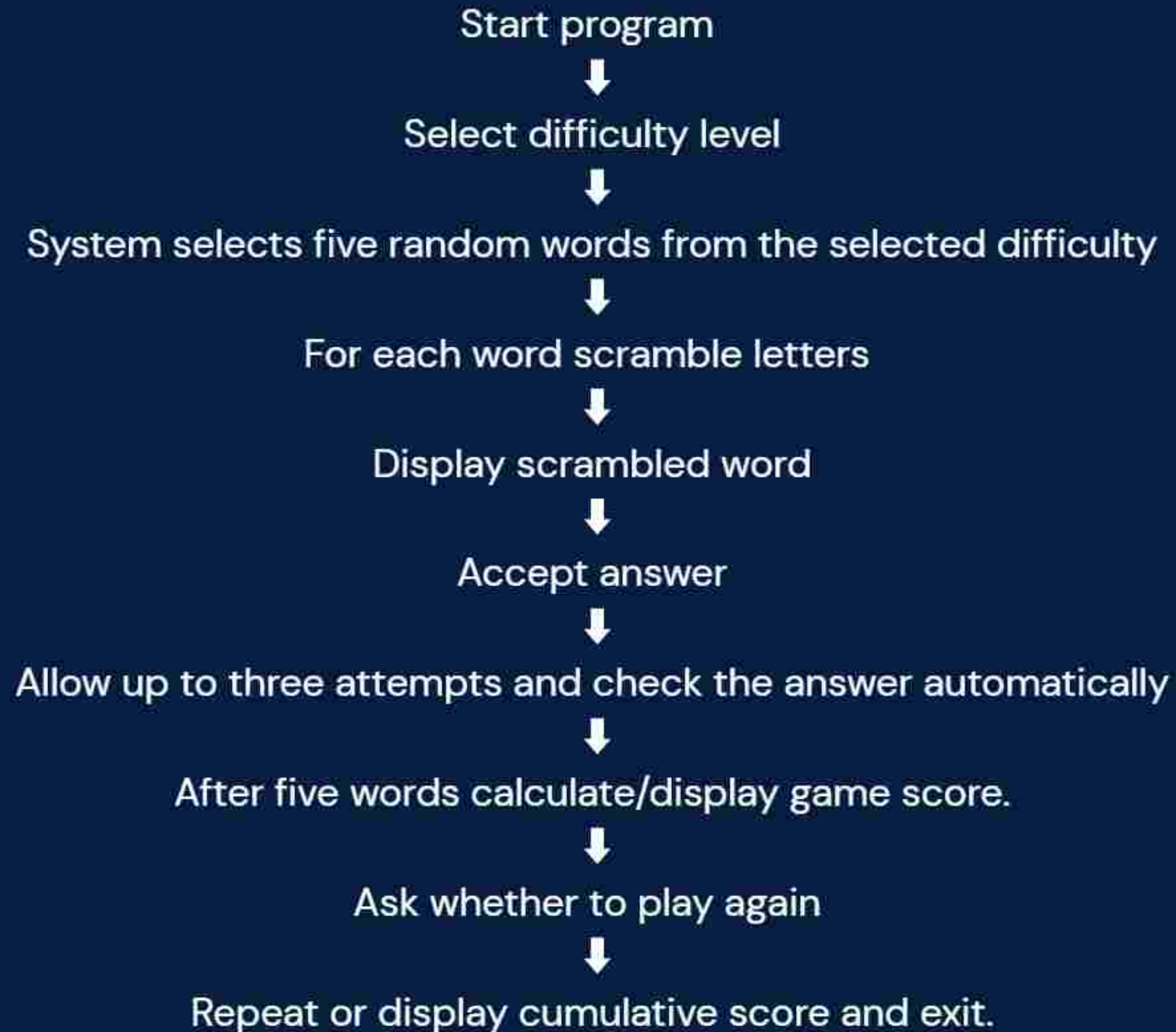
NON-FUNCTIONAL REQUIREMENTS



SYSTEM ARCHITECTURE

The application has a straightforward modular design. The primary controller initiates the game and manages the other modules. Words are provided by the word bank, game logic controls gaming, utility functions deal with interaction and scrambling, input handling verifies text, and score management shows the outcome.

WORKFLOW/PROCESS



USE CASE DIAGRAM



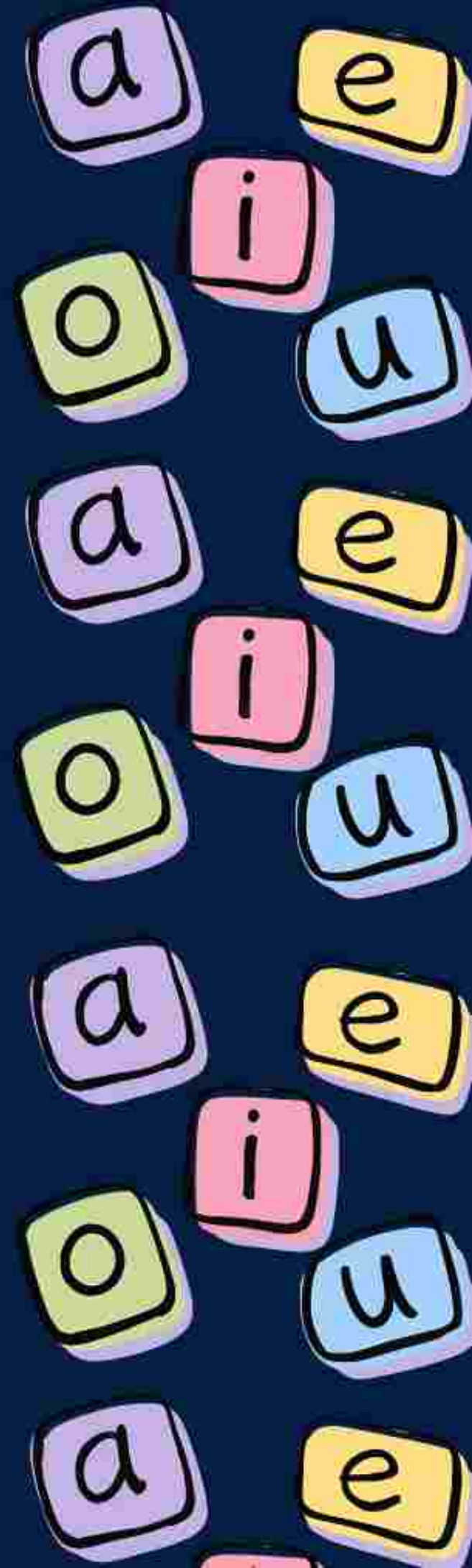


SEQUENCE / COMPONENT DESIGN

- The command line is used to interact with main.py.
- Utility functions and input_handler.py are coordinated by main.py.
- The game asks word_bank.py for words and sends them to game_logic.py.
- The current score is updated and guesses are verified by game_logic.py.
- The cumulative score and game outcome are displayed in score_manager.py. •
- test_game.py independently verifies essential behaviors.

Design Decisions & Rationale

- Python was chosen because of its short syntax and appropriateness for a little interactive game.
- The command-line interface concentrates on fundamental programming ideas and keeps the project lightweight.
- Individual components are easier to maintain and connection is decreased by separate modules.
- Selecting words at random makes them more replayable.
- Word banks with varying levels of difficulty offer a gradual challenge.
- The overhead of dependency management is avoided by using a standard library exclusively.



IMPLEMENTATION DETAILS

- `main.py`: the primary game flow and program entry point.
- `word_bank.py`: words arranged by Easy, Medium, and Hard difficulty are found.
- `game_logic.py`: scoring, scrambling, attempts, answer verification, and random selection.
- `input_handler.py` and `utils.py`: input validation, prompts, and menus.
- `score_manager.py`: display of the game and total score.
- `test_game.py`: data and core rules validation tests.

SCREENSHOTS / RESULTS

Difficulty levels

```
=====
WELCOME TO WORD SCRAMBLE GAME
=====

Choose Difficulty
1. Easy
2. Medium
3. Hard
Enter your choice (1-3): 11
Invalid choice. Please enter 1, 2, or 3.
Enter your choice (1-3): 1
```

Easy

```
=====
WORD SCRAMBLE CHALLENGE
=====

Difficulty: EASY
Unscramble the letters to find the correct word.
You have 3 attempts for each word.

-----

Word 1 of 5
Scrambled word: eakc
Enter your answer: cake
Correct!

-----

Word 2 of 5
Scrambled word: tca
Enter your answer: cat
Correct!

-----

Word 3 of 5
Scrambled word: anri
Enter your answer: rain
Correct!

-----

Word 4 of 5
Scrambled word: obok
Enter your answer: book
Correct!

-----

Word 5 of 5
Scrambled word: zzapi
Enter your answer: zazpi
Wrong answer! Attempts left: 2
Enter your answer: pizza
Correct!
```

Easy result

```
=====
RESULTS
=====

Difficulty: EASY
Score: 5 / 5
Perfect score!

Games played: 1
Total score accrued: 5

Do you want to play again? (y/n): y
```

SCREENSHOTS / RESULTS

Difficulty levels

```
Choose Difficulty
1. Easy
2. Medium
3. Hard
Enter your choice (1-3): 2
```

Medium

```
=====
WORD SCRAMBLE CHALLENGE
=====
Difficulty: MEDIUM
Unscramble the letters to find the correct word.
You have 3 attempts for each word.

-----
Word 1 of 5
Scrambled word: trweri
Enter your answer: winter
Correct!

-----
Word 2 of 5
Scrambled word: leiccyb
Enter your answer: bicycle
Correct!

-----
Word 3 of 5
Scrambled word: uernsm
Enter your answer: summer
Correct!

-----
Word 4 of 5
Scrambled word: shloco
Enter your answer: school
Correct!

-----
Word 5 of 5
Scrambled word: tetblo
Enter your answer: bottle
Correct!
```

Medium result

```
=====
RESULTS
=====
Difficulty: MEDIUM
Score: 5 / 5
Perfect Score!

Games played: 2
Total score accrued: 10
```

SCREENSHOTS / RESULTS

Difficulty levels

```
Do you want to play again? (y/n): y
```

```
Choose Difficulty
```

- 1. Easy
- 2. Medium
- 3. Hard

```
Enter your choice (1-3): 3
```

Hard

WORD SCRAMBLE CHALLENGE

```
Difficulty: HARD
Unscramble the letters to find the correct word.
You have 3 attempts for each word.
```

```
Word 1 of 5
```

```
Scrambled word: aanrstrtue
Enter your answer: architect
Wrong answer! Attempts left: 2
Enter your answer: structure
Wrong answer! Attempts left: 1
Enter your answer: architecture
Wrong answer!
The correct word was: restaurant
```

```
Word 2 of 5
```

```
Scrambled word: nntiaomiga
Enter your answer: imagination
Correct!
```

```
Word 3 of 5
```

```
Scrambled word: kwoenlgde
Enter your answer: knowledge
Correct!
```

```
Word 4 of 5
```

```
Scrambled word: creneseeixp
Enter your answer: expience
Wrong answer! Attempts left: 2
Enter your answer: expence
Wrong answer! Attempts left: 1
Enter your answer: expirence
Wrong answer!
The correct word was: experience
```

```
Word 5 of 5
```

```
Scrambled word: tionaotvin
Enter your answer: motivation
Correct!
```

Hard and final result

RESULTS

```
Difficulty: HARD
Score: 3 / 5
Great job!
```

```
Games played: 3
Total score accrued: 13
```

```
Do you want to play again? (y/n): n
```

GAME OVER

```
Total Games Played: 3
Final Cumulative Score: 13
Thanks for playing!
```

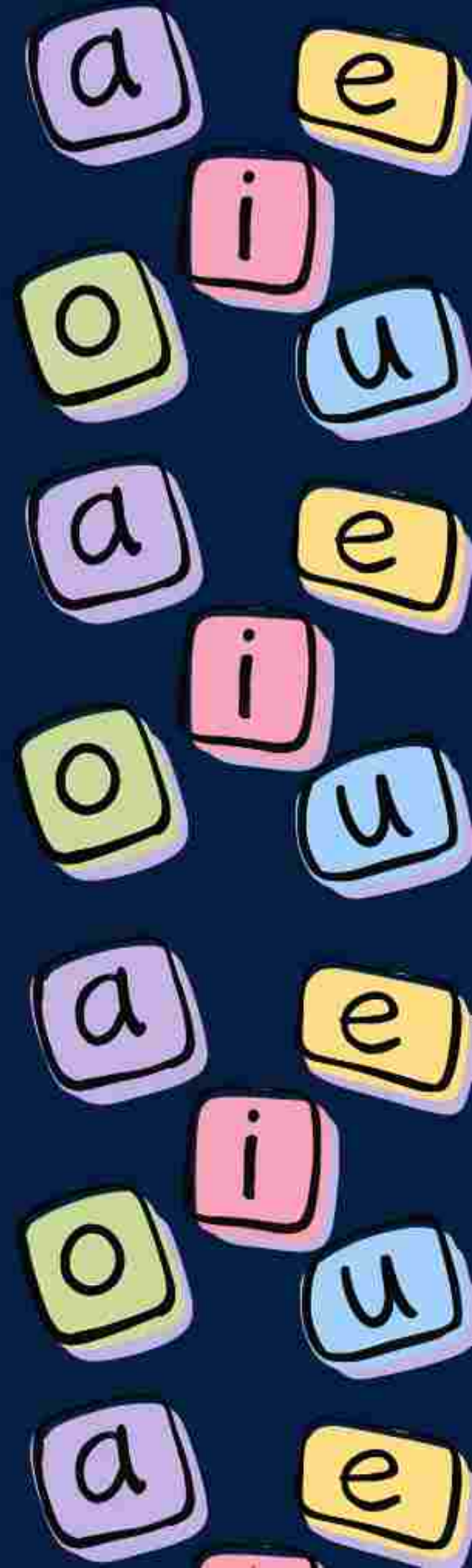


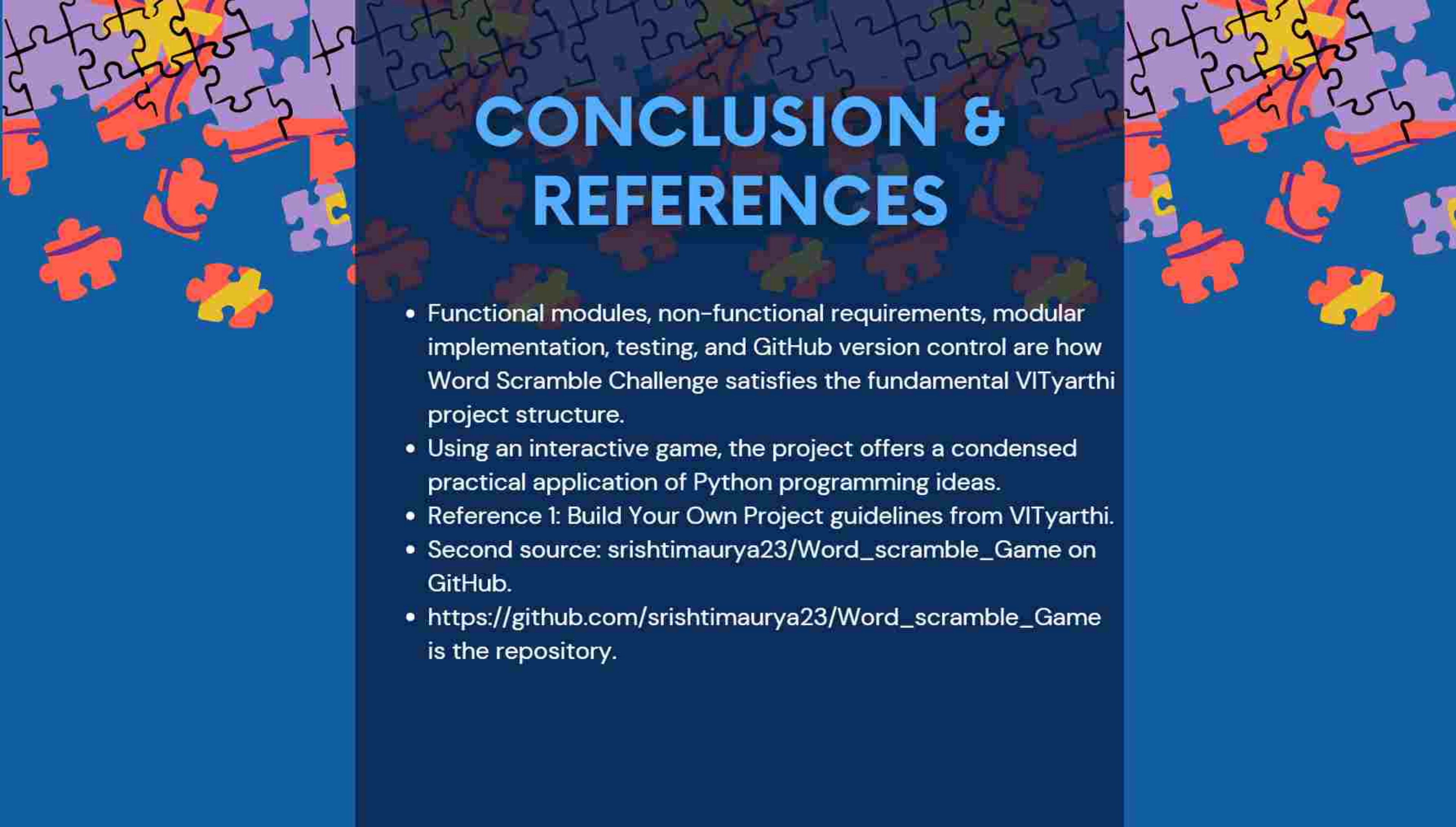
TESTING APPROACH

- Run: `test_game.py` in Python
- Check that the letters in scrambled words match those in the original words.
- Confirm that there are three levels of difficulty.
- Make sure there are at least five words in each difficulty level.
- Confirm that replay validation only accepts y or n.
- Before submission, testing ensures that the fundamental game rules are reliable.

Challenges, Learnings & Future Enhancements

- Designing a clear multi-module flow, verifying user input, and maintaining game state consistency are challenges.
- Acquired knowledge: GitHub-based version control, functions, validation, testing, randomization, and modular Python design.
- Future: include more categories, persistent leaderboards, richer user interfaces, difficulty modification, timer-based scoring, and more automated tests.
- Future improvements should add features gradually while maintaining the existing modular framework.





CONCLUSION & REFERENCES

- Functional modules, non-functional requirements, modular implementation, testing, and GitHub version control are how Word Scramble Challenge satisfies the fundamental VITyarthi project structure.
- Using an interactive game, the project offers a condensed practical application of Python programming ideas.
- Reference 1: Build Your Own Project guidelines from VITyarthi.
- Second source: srishtimaurya23/Word_scramble_Game on GitHub.
- https://github.com/srishtimaurya23/Word_scramble_Game is the repository.

THANK YOU